CORONAVIRUS

Dendrimer nanotherapy for severe COVID-19 attenuates inflammation and neurological injury markers and improves outcomes in a phase2a clinical trial

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Hyperinflammation triggered by SARS-CoV-2 is a major cause of disease severity, with activated macrophages implicated in this response. OP-101, a hydroxyl-polyamidoamine dendrimer–*N*-acetylcysteine conjugate that specifically targets activated macrophages, improves outcomes in preclinical models of systemic inflammation and neuroinflammation. In this multicenter, randomized, double-blind, placebo-controlled, adaptive phase 2a trial, we evaluated safety and preliminary efficacy of OP-101 in patients with severe COVID-19. Twenty-four patients classified as having severe COVID-19 with a baseline World Health Organization seven-point ordinal scale of ≥5 were randomized to receive a single intravenous dose of placebo ($n = 7$ patients) or OP-101 at 2 ($n = 6$), 4 ($n = 6$), or 8 mg/kg ($n = 5$ patients). All study participants received standard of care, including corticosteroids. OP-101 at 4 mg/kg was better than placebo at decreasing inflammatory markers; OP-101 at 4 and 8 mg/kg was better than placebo at reducing neurological injury markers, (neurofilament light chain and glial fibrillary acidic protein). Risk for the composite outcome of mechanical ventilation or death at 30 and 60 days after treatment was 71% (95% CI: 29%, 96%) for placebo and 18% (95% CI: 4%, 43%; $P = 0.021$) for the pooled OP-101 treatment arms. At 60 days, 3 of 7 patients given placebo and 14 of 17 OP-101–treated patients were surviving. No drug-related adverse events were reported. These data show that OP-101 was well tolerated and may have potential to treat systemic inflammation and neuronal injury, reducing morbidity and mortality in hospitalized patients with severe COVID-19.

INTRODUCTION

The coronavirus disease 2019 (COVID-19) pandemic has resulted in >300 million people infected worldwide, with more than 5 million deaths (*1*). Although many patients are mildly symptomatic and recover, up to a third of hospitalized patients may become critically ill (*2*). Since the start of the pandemic, several new variants of the original severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) with higher infectivity continue to spread globally causing substantial morbidity and mortality (*3*). The pathophysiology of severe disease is independent of the variant and is characterized by an exuberant cytokine response syndrome or hyperinflammation that can cause severe pulmonary injury and acute respiratory distress syndrome (ARDS), multiorgan dysfunction, and death. Monocytes and activated tissue macrophages likely play an important role in the hyperinflammatory response seen in these severely affected patients (*4*, *5*) and are also well known to play a role in the pathophysiology of ARDS (*6*).

Apart from the pulmonary and systemic effects, neuropsychiatric symptoms are commonly associated with COVID-19, and presentations have ranged from delirium and agitation to encephalopathy and stroke (*7*, *8*). Recent studies showed that increases in brain injury markers such as serum neurofilament light chain (NfL) and glial fibrillary acidic protein (GFAP) correlated with unfavorable outcomes, and treatment with steroids or monoclonal antibodies did not have an effect on these markers (*9*, *10*). Given that direct invasion of the central nervous system (CNS) by the virus has not been clearly demonstrated, it is likely that the neurological effects are related to systemic inflammation driving neuroinflammation and CNS dysfunction (*11*).

OP-101 is a nanotherapeutic compound that has previously been evaluated in several animal models of inflammatory disease including models of brain and systemic inflammation (*12*–*20*). OP-101 also has been evaluated in phase 1 studies in healthy human volunteers (NCT03500627, NCT04321980). OP-101 is a covalent conjugate of generation-4 hydroxyl terminated poly(amidoamine) dendrimer (~4 nm in size) and *N*-acetyl cysteine (NAC), enabled by a disulfide linker that readily releases the drug through the action of intracellular glutathione (*21*, *22*). Preclinical studies have shown that intravenously delivered hydroxyl polyamidoamine (PAMAM) dendrimers and their conjugates selectively localize in activated microglia/macrophages in the CNS in small and large animal models and deliver drugs to the site of injury producing positive therapeutic outcomes (*14*, *18*–*20*). Relatively minimal cellular uptake is seen in healthy animals and in normal macrophages (*12*, *16*–*18*). OP-101 delivers NAC inside activated macrophages and microglia, resulting in superior anti-inflammatory and antioxidant effects than seen with high free NAC doses (*12*, *14*, *18*, *20*). Our phase 2a clinical study aims to describe the pharmacokinetics and tolerability of OP-101, its impact on markers of inflammation and brain injury, and offers preliminary insights about its impact on morbidity and mortality in patients with severe COVID-19.

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RESULTS

Phase 2a clinical study design and patient enrollment

This phase 2a clinical study [OP-101 to arrest hyperinflammation in COVID-19 patients (PRANA)] was designed as a multicenter, randomized, double-blind, placebo-controlled, single sequential, ascending-dose study for evaluating safety and tolerability of OP-101 and to obtain preliminary insights about efficacy in patients with severe COVID-19. Twenty-four patients with a World Health Organization seven-point ordinal scale (WHO 7OS) (*23*) (table S1) score of ≥5 (classified as severe COVID-19) were enrolled at five U.S. centers, between August 2020 and March 2021. Enrollment was done in three ascending dose cohorts of eight patients each (3:1 enrollment ratio of OP-101 to placebo) as outlined in Fig. 1. After completion of day 3 assessments at each dose level, the independent data monitoring committee (IDMC) reviewed clinical and laboratory data and determined whether it was safe to proceed to the next dose level. No drug-related concerns were noted by the IDMC, and dose escalation occurred as planned after each review was completed. In the final dose cohort, because of a change in a patient's ordinal score before randomization, an imbalance occurred resulting in five patients in the 8 mg/kg arm with a total of seven patients in the placebo group.

Baseline demographic and clinical characteristics of the study sample are shown in Table 1. Demographic characteristics and comorbidities were reasonably well balanced among groups. All patients had a WHO 7OS score of ≥5, and the median WHO 7OS score was 5 in all groups. Remdesivir use was common, and all subjects were treated with corticosteroids as per standard of care during the enrollment period (*24*). The most common modality of respiratory support was high-flow nasal cannula (HFNC) as shown in Table 1. There were more men enrolled in the study than women (17 versus 7), which is consistent with COVID-19–associated hospitalization rates as men are at higher risk of increased disease severity (*25*). Diabetes as a comorbidity was seen in 14 of 24 patients. Patients with an age > 45 and the presence of comorbidities such as diabetes mellitus and obesity have been shown to have a higher risk for increased severity of illness (*26*). The patients in this study were generally representative of the patient population hospitalized with severe COVID-19.

Predose C-reactive protein (CRP), absolute lymphocyte count (ALC), and neutrophil-to-lymphocyte ratio (NLR) were similar between the groups (P = 0.17, 0.27, and 0.2, respectively, based on a Wilcoxon rank sum test) (Table 1). An ALC of $<1 \times 10^9$/liter (*27*), CRP of ≥40 mg/liter (*28*), or NLR of >6.82 (*29*) has been shown to be independently associated with severe disease and increased mortality. Seventeen of 24 patients had lymphopenia, and 5 patients had severe lymphopenia ($<0.5 \times 10^9$/liter). Predose CRP was available for 21 of 24 patients. Only three patients had a predose or 2 hours after infusion CRP less than 40 mg/liter, one each in the placebo, 4 mg/kg, and 8 mg/kg groups (fig. S1). Elevated CRP and NLR and lymphopenia were seen in most patients indicative of more severe disease that matches the clinical presentation of a WHO 7OS of ≥5 at the time of enrollment in the study (Table 1). Inflammatory markers and neurological injury markers were measured using frozen samples at a later time point and were compared between groups. There were no statistically significant differences in predose inflammatory markers or neurological injury markers between the different groups except for serum interleukin-6 (IL-6) that was higher in the placebo group (P = 0.013, table S2A). Given that predose samples were

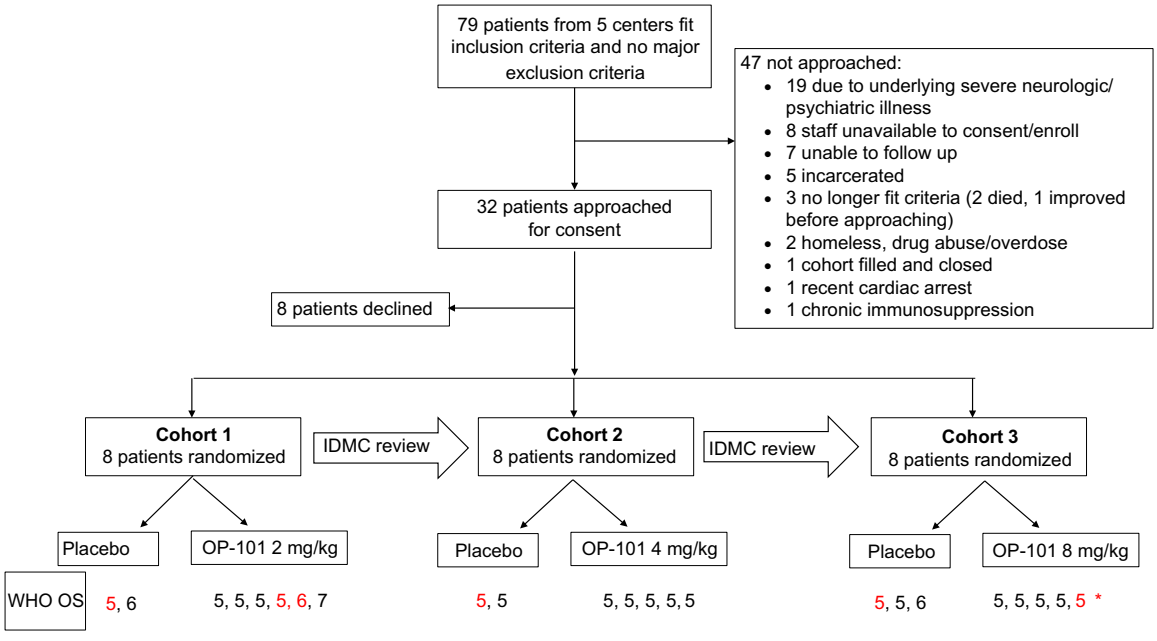


Fig. 1. CONSORT diagram for the phase 2a clinical study. Flow chart shows patient enrollment in this phase 2a multicenter, randomized, double-blind, placebo-controlled, single sequential, ascending-dose study for evaluating safety and tolerability of OP-101 in patients with severe COVID-19. Patients were enrolled at five clinical institutions across the United States. Among the 32 patients who met inclusion and exclusion criteria and were approached for consent, 24 patients consented for the trial and were enrolled. Patients were enrolled in three separate and sequential, ascending dose cohorts (2, 4, and 8 mg/kg) with an enrollment ratio of 3:1 OP-101 to placebo. Dose escalation from one cohort to the next higher dose cohort was determined by the independent data monitoring committee (IDMC) after review of safety and tolerability through day 3. Numbers at the bottom indicate the WHO 7 point ordinal scale (WHO 7OS), a measure of COVID-19 severity, for each patient upon enrollment. The initial ordinal scale of patients who died is indicated in red. * indicates the patient whose care goals were changed to comfort care and support was withdrawn.

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Table 1. Baseline characteristics of patients enrolled in each group. MV, mechanical ventilation; HFNC, high-flow nasal cannula; BiPAP, bilevel positive airway pressure; CPAP, continous positive airway pressure. Baseline ALC and NLR values of patients who died are marked in red.

Patient characteristics	Placebo ($n = 7$)	OP-101 2 mg/kg ($n = 6$)	OP-101 4 mg/kg ($n = 6$)	OP-101 8 mg/kg ($n = 5$)
Age (years) and median (min, max)	61 (43, 76)	61 (52, 67)	57.5 (45, 66)	61 (59, 86)
Sex (M/F)	5/2	5/1	4/2	3/2
Ethnicity				
Black	2	0	1	2
Hispanic or Latino	5	5	3	2
Non-Hispanic White	0	1	2	1
BMI (kg/m^2) and mean (min, max)	32.2 (24.2, 47.7)	28.54 (24.3, 32.9)	31.4 (23.9, 40.1)	28.9 (23.3, 33.8)
Predose WHO 7OS at randomization and median (min, max)	5.0 (5, 6)	5.0 (5, 7)	5.0 (5, 5)	5.0 (5, 5)
Maximal respiratory support on day of infusion	HFNC (4)	HFNC (4)	HFNC (4)	HFNC (3)
	BiPAP (1)	MV (2)	MV (2)	BiPAP (1)
	MV (2)			CPAP (1)
Absolute Lymphocyte Count (10^9/L) (ALC) Median [values, patients who died are indicated in red]	0.8 [0.9, 0.8, 1.8, 0.4, 0.4, 1.4, 0.8]	0.8 [0.3, 0.6, 0.9, 1.0, 0.7, 0.8]	0.5 [0.2, 0.9, 0.3, 0.5, 0.5, 1.1]	1.0 [1, 1.3, 0.6, 1.5, 0.9]
Neutrophil/Lymphocyte Ratio (NLR); Median, [values, patients who died are indicated in red]	17.1 [14.2, 17.1, 9.5, 17.8, 43, 8.8, 19.8]	8.6 [26.5, 10, 8, 9.1, 3.9, 6.9]	15.6 [94.4, 5.7, 29.7, 5.7, 19.6, 11.6]	9.2 [5.5, 8.5, 11.2, 9.2, 13.6]
C-reactive protein (mg/liter) and median [min, max]	150.5 [64, 349] ($n = 6$); 22.9*	192 [112, 257.8] ($n = 5$); 47.9*	104.8 [27.5, 216] ($n = 6$)	90.6 [29, 143.1] ($n = 4$)
†COVID-related medications				
Remdesivir	4	4	6	4
Corticosteroids	7	6	6	5
Antithrombotic meds	6	6	6	5
Comorbidities				
Diabetes mellitus	3	4	4	3
Hypertension	1	3	3	4
Others	0	0	1‡	1§

*These represent values at 2-hour after infusion that are similar to baseline values. These patients did not have a predose measurement of CRP. †These medications were administered during their hospital course. ‡Patient had history of basal cell carcinoma. §Patient had history of chronic liver failure from hepatitis C virus infection and emphysema.

obtained within 24 hours before drug infusion, we also compared IL-6 concentrations at 2 hours after infusion between the groups, when we would not expect a response to treatment. There were no significant differences between the groups for all markers at this time point (table S2B).

OP-101 demonstrates dose-proportional pharmacokinetics

Serum concentrations of OP-101 were measured at 2, 12, and 24 hours after intravenous administration of OP-101 at 2, 4, or 8 mg/kg, or placebo. We have previously shown that the hydroxyl dendrimer is rapidly cleared from the circulation through the urine and that the selective dendrimer uptake by activated microglia and macrophages is dependent on the severity of injury and degree of inflammation (*12*, *16*). As shown in Fig. 2 and Table 2, the pharmacokinetics and plasma exposure of this nanotherapeutic were proportional to the dose administered with an approximately twofold increase in area under the curve (AUC) seen with doubling of the dose [AUC mean (min, max) was 84.7 (39.2, 130.2) for 2 mg/kg; 178.2 (78.07, 227.3) for 4 mg/kg; and 363.8 (183.8, 641.2) for 8 mg/kg].

OP-101 treatment reduces serum CRP concentrations

CRP concentrations in serum were measured predose and at 2, 12, 24 hours, 2, 4, and 7 days after infusion of OP-101. The change in CRP from predose concentrations for all patients in each of the groups is depicted in Fig. 3A. In general, CRP concentrations remained similar to predose concentrations in the placebo group, whereas a steady and sustained decrease was noted for the OP-101–treated patients. Previous studies have shown that a >50% drop in CRP from

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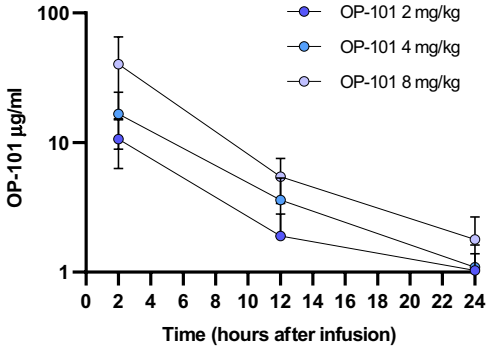


Fig. 2. Pharmacokinetics of OP-101. OP-101 concentrations were measured in patient serum collected at 2, 12, and 24 hours after intravenous administration of one dose of OP-101 at 2, 4, or 8 mg/kg. Mean and standard deviation at each time point is shown for all the doses. A dose-proportional pharmacokinetic profile was obtained for OP-101. OP-101 was cleared rapidly with most of the drug being eliminated by about 24 hours after infusion.

baseline in the first 72 hours after starting steroids was associated with a better outcome (*30*). In our study, the timing between starting steroids and OP-101 treatment was variable. However, patients treated with OP-101 appeared to have a more rapid and sustained decrease in CRP, indicating that OP-101 may have additional benefits over steroids. Although not reaching significance, we noted that by day 3 (48 hours after infusion), 75% of patients treated with OP-101 (12 of 16) had a ≥50% decrease in CRP concentrations from predose concentrations compared to 29% (2 of 7) of patients who received placebo (Fig. 3B). The rate of change in CRP (slope) after infusion was compared between the groups. The median decrease in CRP from baseline was <5% per day in the placebo group, whereas treatment with OP-101 resulted in a median decrease of >20% per day for all doses with a significant decrease in the rate of change noted for the 4 mg/kg dose when compared to placebo [$P = 0.032$ for placebo versus 4 mg/kg OP-101) (Fig. 3C and table S3).

Treatment with OP-101 attenuates an increase in neurological injury markers

We measured NfL with an immunoassay and GFAP and total Tau protein with the Quanterix 4-plex NeuroA assay in frozen serum samples collected at predose (baseline), 2, 12, and 24 hours after infusion and days 3, 5, 8, 15, 22, and 30 after infusion. NfL is a cytoskeletal intermediate filament protein that can arise from the central and peripheral nervous system (*31*). GFAP is primarily expressed by astrocytes, whereas Tau is mostly expressed by neurons in the CNS (*32*). We found that patients in the placebo group had a greater increase in serum NfL over time when compared to patients treated with OP-101 (Fig. 4A). The rate of change (slope) was higher for the placebo group with a median increase of 14.4% per day compared to the OP-101 dosing groups with a significant difference noted for the two highest doses [median change was +3.3% for 4 mg/kg group ($P = 0.032$); +0.2% for the 8 mg/kg group ($P = 0.004$)] (Fig. 4B and table S3). Similarly, the rate of increase for GFAP was significantly higher for the placebo group compared to each OP-101 treatment group (Fig. 4C, fig. S2, and table S3). In our study, although serum Tau protein increased over time for the placebo group when compared to the OP-101–treated groups, a significant difference between the groups was not noted (Fig. 4D, fig. S3, and table S3). This attenuation in the increase in neurological injury markers after OP-101 treatment appeared to last until day 30 after treatment (fig. S7, A to C). These data indicate that treatment with OP-101 prevented the increase in brain injury markers that has been reported previously in patients with severe COVID-19 despite the use of corticosteroids.

Treatment with OP-101 reduces proinflammatory markers

The inflammatory markers tumor necrosis factor–α (TNF-α), IL-6, IL-8, IL-22, and IL-10 were measured over time in available serum samples from the patients in the study, as part of a multiplex inflammatory panel. The rate of change after treatment for TNF-α, IL-6, and IL-8 serum concentrations for each patient is depicted in Fig. 5 (A to C). Patients given placebo had a median daily increase of +7% for IL-6, +7.7% for TNF-α, and +10% for IL-8. Treatment with 4 mg/kg OP-101 was associated with a significant decrease over time [slope of trajectory depicted as rate of change per day in Fig. 5 (A to C)] when compared to placebo for TNF-α ($P = 0.016$), IL-6 ($P = 0.016$), and IL-8 ($P = 0.025$) (table S3). Individual patient trajectories for IL-6, TNF-α, and IL-8 are provided in fig. S4. No differences between groups were noted for IL-22 and IL-10 (table S3).

Treatment with OP-101 leads to improved clinical outcomes

Treatment with OP-101 was associated with an improvement in survival. Survival was 43% [95% confidence interval (CI): 10%, 82%)] for the placebo group (three of seven patients), 67% (95% CI: 22%, 4%) for the OP-101 2 mg/kg group (four of six patients survived), 100% (95% CI: 54%, 100%) for the 4 mg/kg group (six of six), and 80% (95% CI: 28%, 99.5%) for the 8 mg/kg group (four of five) (Fig. 6).

Among the three surviving patients in the placebo group, one remained hospitalized on mechanical ventilation until the end of the study (day 60). None of the surviving OP-101–treated patients were on mechanical ventilation on follow-up at day 30 and day 60. The risk of 30-day outcome of mechanical ventilation or death for the placebo group was 71% (95% CI: 29%, 96%) and 18% (95% CI: 4%, 43%) for the combined OP-101 treatment arms. The relative risk reduction associated with OP-101 was 75% (= 1 − [0.18/0.71]; Fisher's exact $P = 0.021$). Whereas the counterfactual interpretation of this reduction is limited because of the small sample size, these findings in which treatment arms were pooled offer a preliminary glimpse toward putative efficacy estimates. An assessment of WHO 7OS over 30 days after OP-101 treatment indicated that there was a tendency for more rapid improvement in the scores in the OP-101–treated patients (fig. S5). However, because of the small number of patients in the placebo group at later time points, no statistical analysis was done.

Safety and tolerability of OP-101

Safety assessments were done after a single bolus intravenous administration of OP-101 or placebo (saline) on days 1, 2, 3, 8, 15, 22, and 30 by investigators who were blinded to treatment groups. Given that all patients had severe COVID-19 and were critically ill, treatment-emergent adverse events (TEAEs) were reported in most patients as expected for this population (Table 3). All TEAEs (reported in half of patients) were determined by the investigators to be caused by COVID-19. Of the patients who received any dose of OP-101, 12 of 17 (70.5%) had TEAEs of whom 8 of 17 (47%) had serious TEAEs, whereas in the placebo group, 5 of 7 (71.4%) patients had TEAEs, and 4 of 7 (57.1%) had serious TEAEs. There were no significant differences in adverse events (AEs) between the placebo and OP-101–treated groups (Table 3).

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Table 2. Pharmacokinetic parameters: AUC last, T_{max}, and C_{max} of OP-101. AUC_{last} indicates the last measurable concentration that was used to calculate the area under the curve (AUC). Three patients had a plasma OP-101 concentration at ~24 hours that was below the limit of detection. For these patients, the last measurable concentration obtained at 12 hours was used to derive the AUC. One patient (ID 301) in the OP-101 4 mg/kg arm had only one sample for PK measurements that was collected; therefore, PK parameters could not be calculated. T_{max}, time for peak concentration; T_{last}, time at last measurable concentration; C_{max}, peak concentration; C_{last}, last measurable concentration.

OP-101 dose	Subject ID	AUC_{last} (hour*μg/ml)	T_{max} (hours)	T_{last} (hours)	C_{max} (μg/ml)	C_{last} (μg/ml)
2 mg/kg	201	113.6867	2.500	24.50	11.40	1.440
2 mg/kg	202	109.4630	2.067	24.22	12.50	0.829
2 mg/kg	203	68.4325	2.400	12.70	9.69	1.340
2 mg/kg	204	39.1860	2.500	12.20	5.83	0.747
2 mg/kg	205	130.1737	1.917	24.13	17.70	0.819
2 mg/kg	206	47.4268	2.050	11.95	6.67	1.530
4 mg/kg	301	–	2.883	–	9.14	–
4 mg/kg	302	227.2825	2.283	24.22	25.40	1.880
4 mg/kg	303	218.0162	2.533	24.53	25.20	1.390
4 mg/kg	304	204.7480	10.617	32.58	17.10	0.836
4 mg/kg	305	163.0528	5.417	27.50	16.10	0.692
4 mg/kg	306	78.0668	7.050	27.67	6.93	0.655
8 mg/kg	401	453.5690	3.333	23.40	37.00	2.060
8 mg/kg	402	183.7683	2.783	24.67	23.80	0.626
8 mg/kg	403	205.5119	2.750	24.20	23.50	1.310
8 mg/kg	404	641.1964	2.850	25.58	83.70	2.020
8 mg/kg	405	334.9923	4.667	25.90	33.40	2.940

Given that OP-101 is primarily eliminated renally, we also measured urine concentrations of kidney injury molecule-1 (KIM-1) and neutrophil gelatinase-associated lipocalin (NGAL) as markers of kidney tubule injury. There was no significant difference in KIM-1 and NGAL between patients who were treated with placebo or all doses of OP-101 (fig. S6).

DISCUSSION

In this multicenter, randomized, double-blind, placebo-controlled phase 2a trial, we evaluated OP-101, a hydroxyl dendrimer nanotherapeutic, in patients with severe COVID-19. This first-in-human pilot trial demonstrated that OP-101 was well tolerated in critically ill COVID-19 patients and showed promising preliminary efficacy with an improvement in the composite clinical outcome of mechanical ventilation or death up to day 60, as well as attenuation of inflammation and neurological injury markers. The primary endpoint of demonstrating safety and tolerability of this nanotherapy in critically ill patients was met with no drug-related AEs noted.

The pharmacokinetics of OP-101 appear to follow the pharmacokinetics of the hydroxyl PAMAM dendrimer (size, ~4 nm). Previous studies have shown that the hydroxyl PAMAM dendrimer used here has a short circulation time with selective localization in activated macrophages, and dendrimer that is not retained intracellularly is eliminated intact in the urine by ~24 hours (*12*, *33*). OP-101 is eliminated primarily through the kidney with a decrease in plasma concentration to <2 μg/ml by 24 hours for all doses. Despite a relatively short circulation time, it is possible that the targeted and sustained intracellular localization and release in activated macrophages potentially enable prolonged efficacy and sustained pharmacodynamic effects due to a change in macrophage/microglial phenotype enabling tissue repair as shown in preclinical studies (*16*, *18*–*20*).

Hyperinflammation and oxidative stress have been demonstrated in patients with severe COVID-19 (*34*). The active agent in OP-101 is NAC that has both antioxidant and anti-inflammatory properties (*35*). NAC is an inhibitor of nuclear factor κB (NF-κB) and can prevent replication of RNA viruses due to its action on NF-κB [reviewed in (*36*)]. Although we did not test for this here, NF-κB inhibition by NAC may also result in antiviral effects and may prevent replication of SARS-CoV-2. Reactive oxygen species leading to oxidative injury have been linked to worsening lung injury in ARDS. NAC has been shown to be effective in some studies in patients with ARDS (*37*) and in reducing mortality in patients with COVID-19 pneumonia in a small, observational, retrospective study (*38*). NAC is usually used in very high doses because of poor bioavailability due to protein binding of the sulfhydryl groups (*39*). Dendrimer-NAC has been shown to be more efficacious and potent than the free-drug NAC at much lower doses, because it specifically targets activated macrophages at the site of injury increasing local bioavailability (*18*). In this study, preliminary efficacy was seen with a single dose of OP-101 at a fraction of the dose of NAC (~1.6 mg/kg of NAC) typically used for other clinical indications (total dose of >100 mg/kg). This indicates that targeted delivery of OP-101 is probably responsible for the promising efficacy seen here.

We observed a decrease in serum CRP with OP-101 treatment. These differences in CRP may be both due to a direct effect of OP-101 on macrophages and an indirect effect of a decrease in proinflammatory cytokines. Although hepatocytes in response to proinflammatory

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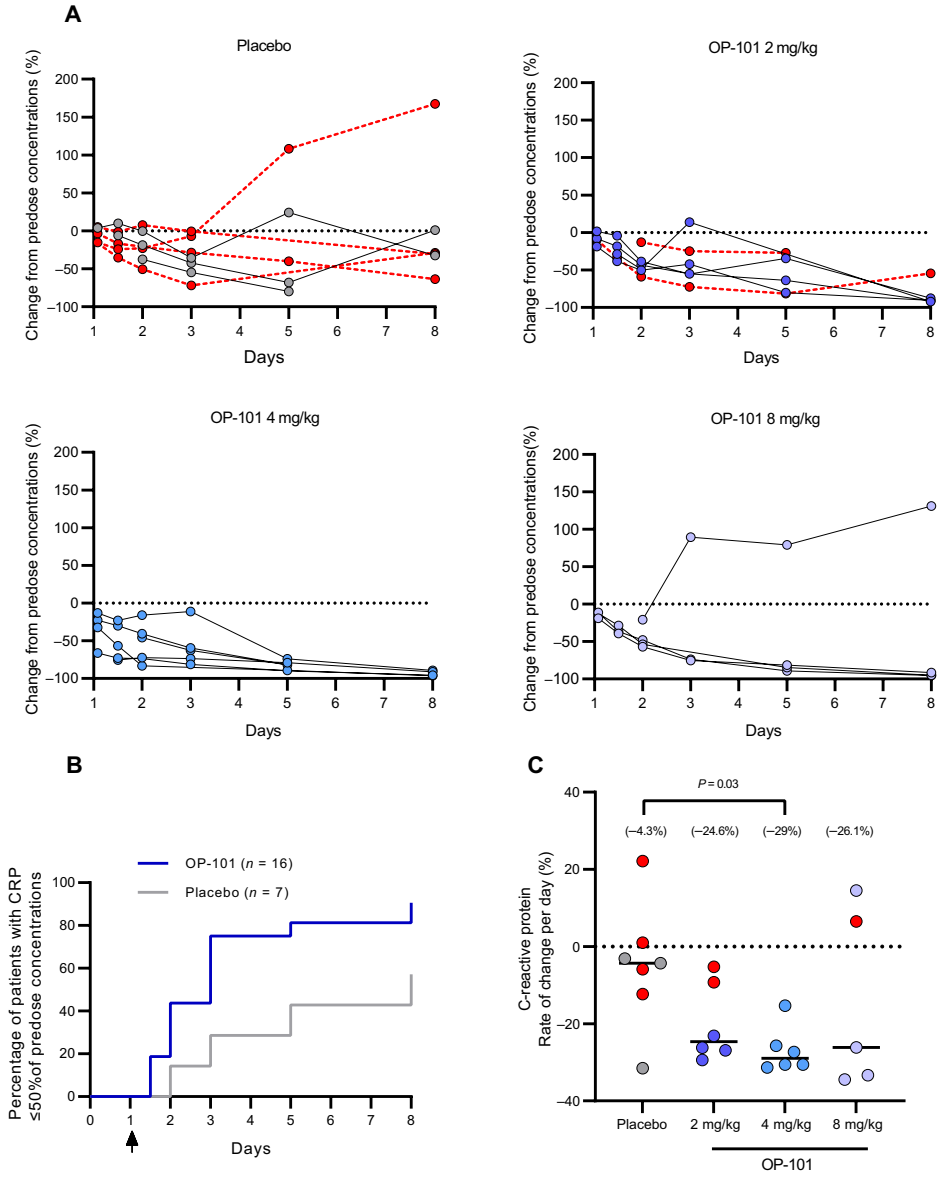


Fig. 3. CRP serum concentrations over time after OP-101 treatment. (**A**) Shown is the percent change in plasma C-reactive protein (CRP) from predose concentrations until day 8 (day 7 after infusion) for individual patients in each treatment cohort (2, 4, or 8 mg/kg OP-101) or the placebo group. Day of infusion is depicted as day 1. Because CRP serum concentrations were measured at different local laboratories, individual data were normalized to their predose values. Patients who died are depicted in red. One patient in the placebo group and one patient in the 2 mg/kg OP-101 treatment group did not have predose CRP concentrations measured; for these two patients, the 2-hour value was taken as baseline as we would not expect response to treatment in 2 hours ($n = 7$ for placebo, $n = 6$ for OP-101 2 mg/kg, $n = 6$ for OP-101 4 mg/kg, and $n = 4$ for OP-101 8 mg/kg). The patient who died in the 8 mg/kg cohort did not have CRP values measured predose or within 24 hours of infusion and therefore was not included in this graph. (**B**) The proportion of patients who achieved CRP serum concentrations less than 50% of baseline between day 1.5 (12 hours after infusion) and day 8 is depicted as a Kaplan-Meier curve for the pooled OP-101–treated patient groups (blue) and the placebo group (gray). Day 1 is day of infusion (indicated by an arrow). (**C**) Rate of change in CRP (slope) is expressed as percent change per day for each participant after treatment until day 8 after infusion. The overall trajectories in CRP serum concentrations after treatment for each individual is summarized as a slope (i.e., assumption of linear change) by regressing all available CRP values on time as days. Therefore, the slope for each patient represents the average rate of change in CRP per day, after infusion. Wilcoxon rank sum test was used with placebo as reference to test differences across treatment arms. Horizontal bar indicates median rate of change for that group (median value in parentheses). A greater rate of decrease in CRP was noted for the OP-101 treatment groups with a significant difference noted for the 4 mg/kg group when compared to placebo ($P = 0.03$). Patients who died are depicted in red.

cytokines are the most common site for CRP production, monocytes and macrophages including alveolar macrophages have been shown to produce CRP (*40*). It is possible that inhibition of production of CRP by macrophages and by the liver due to a decrease in proinflammatory cytokines by macrophages resulted in the decrease in CRP trajectories noted with OP-101 treatment.

Neurological and psychiatric complications occur frequently after COVID-19 hospitalization, especially after severe COVID-19 (*41*). Brain injury markers such as NfL, GFAP, and Tau are elevated in hospitalized patients with COVID-19 (*42*) and are associated with worse outcomes and increased mortality (*9*, *10*, *43*, *44*). We found an increase in serum NfL and GFAP over time in patients given placebo, which was attenuated in OP-101–treated patients. Of note, NfL lacks specificity for the CNS and has also been shown to be elevated in patients with peripheral neuropathies and chronic inflammatory demyelinating polyneuropathy (*45*). Although we did not have cerebrospinal fluid (CSF) samples available to assess CNS NfL, strong correlations have been shown between CSF and blood concentrations of NfL, suggesting a unifying pathophysiological process (*46*). It is possible that an increase in NfL may be due to peripheral neuropathy in patients with severe COVID-19. Peripheral nerve injury has been described after COVID-19 and may be associated with macrophage activation and immune dysregulation (*47*). In this preliminary study, these markers were exploratory endpoints, and we did not conduct neuropsychiatic evaluations or evaluate patients for peripheral neuropathy to assess clinical improvement with OP-101 treatment. These outcomes will need to be pursued further in larger clinical trials. Persistent, long-term neurological symptoms occur commonly after SARS-CoV-2 infection, with patients often experiencing cognitive symptoms and "brain fog" (*48*). Microglial activation and transcriptome changes have been described in postmortem specimens from patients with severe COVID-19 in the absence of active SARS-CoV-2 infection (*11*). Increased inflammation and immune dysregulation may persist long after viral clearance (*49*), which may contribute to microglial activation and cognitive dysfunction (*50*).

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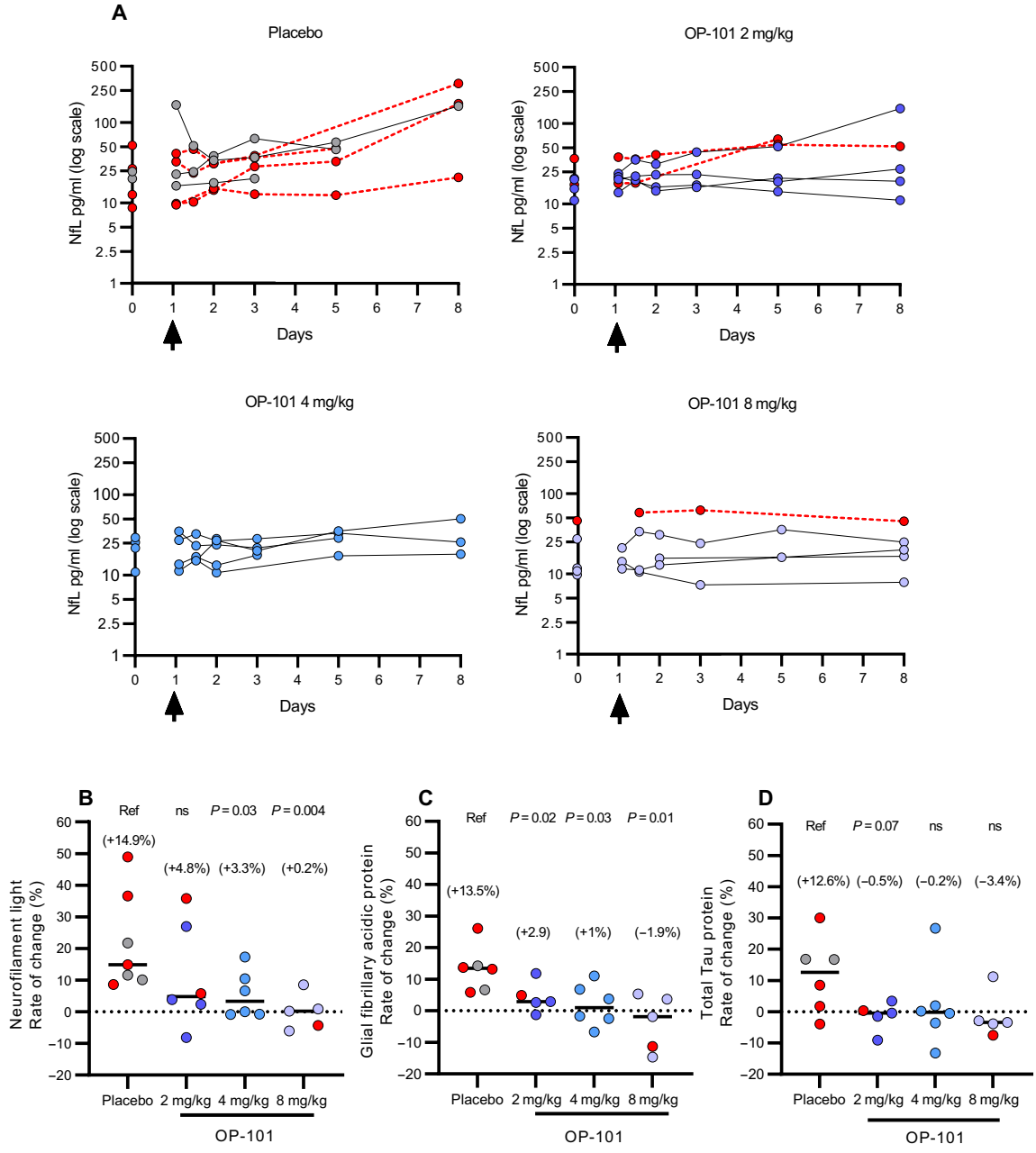


Fig. 4. Change in neurological injury markers over time after OP-101 treatment. (**A**) Trajectories of serum NfL values over time after treatment are shown. Patients who died are marked in red. Day of infusion is depicted as day 1 (arrow). "0" indicates predose serum concentration of NfL. In general, patients who received placebo showed an increase in serum NfL over time, and this increase was not noted for patients treated with the three doses of OP-101 (2, 4, and 8 mg/kg). (**B** to **D**). The average rate of change per day (slope) for each analyte in serum expressed as percent change after treatment is shown for each individual for the neurological injury markers NfL, GFAP, and Tau protein. Patients who died are depicted in red. Horizontal bar indicates the median rate of change for that treatment group (median value shown in parentheses). Serum NfLand GFAP showed a significant increase in rate of change (slope) for the placebo group but showed minimal change over time for the three OP-101 treatment groups. Total Tau protein (D) also showed an increase for the placebo group but not for the OP-101 treatment groups, but this did not reach significance. Two patients (one in the placebo group and one in the OP-101 2 mg/kg cohort) did not have adequate serum samples aliquoted for any of the Quanterix assays (GFAP and Total Tau). *P* values are based on the Wilcoxon rank sum test. The number of patients for each time point for all assays is provided in table S4. Ref, reference; ns, not significant.

SARS-CoV-2 infection results in complex and incompletely understood interactions with the immune system. An early maladaptive increase in proinflammatory cytokines, such as IL-1β, IL-6, CXCL8, and TNF-α has been reported (*51*). Peripheral blood from patients with COVID-19 in the intensive care unit (ICU) demonstrated expansion of IL-6–producing $CD14^{+}CD16^{+}$ monocytes (*52*). However, there is reason to believe that circulating monocytes may not be the predominant contributor to cytokine storm. Single-cell RNA sequencing of peripheral blood mononuclear cells demonstrated that peripheral blood monocytes did not express substantial amounts of proinflammatory cytokines (*53*). The lack of robust cytokine production from blood mononuclear cells and evidence of peripheral

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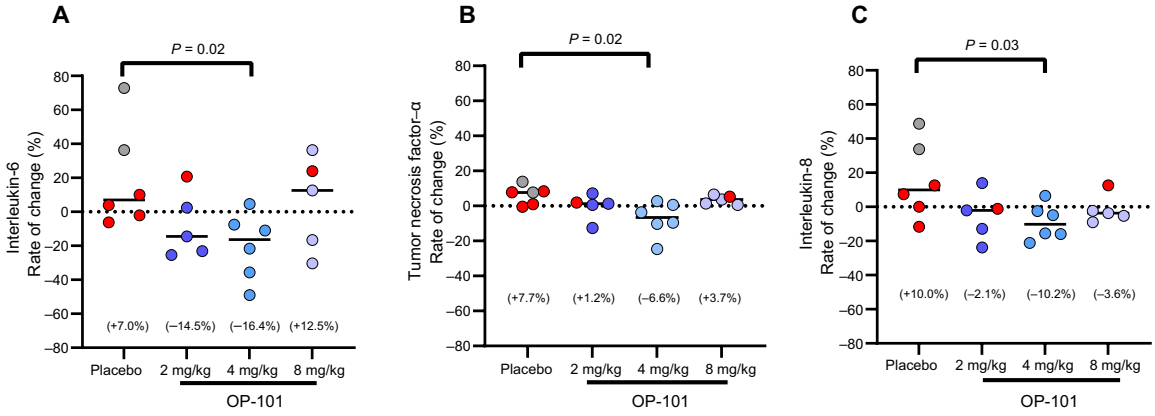


Fig. 5. Change in inflammatory markers over time after OP-101 treatment. (**A** to **C**) Shown is the change in inflammatory markers over time (slope) expressed as rate of change (%) for each patient after treatment with OP-101 at 2, 4, or 8 mg/kg. The average rate of change per day (slope) for IL-6 (A), TNF-α (B), and IL-8 (C), expressed as percent change after treatment, is shown for each individual. Patients who died are indicated in red. A significant decrease over time was noted for the patients treated with 4 mg/kg OP-101 when compared to placebo for all three cytokines. Horizontal bar indicates median change, and median values are shown in parentheses. *P* values are based on Wilcoxon rank sum test.

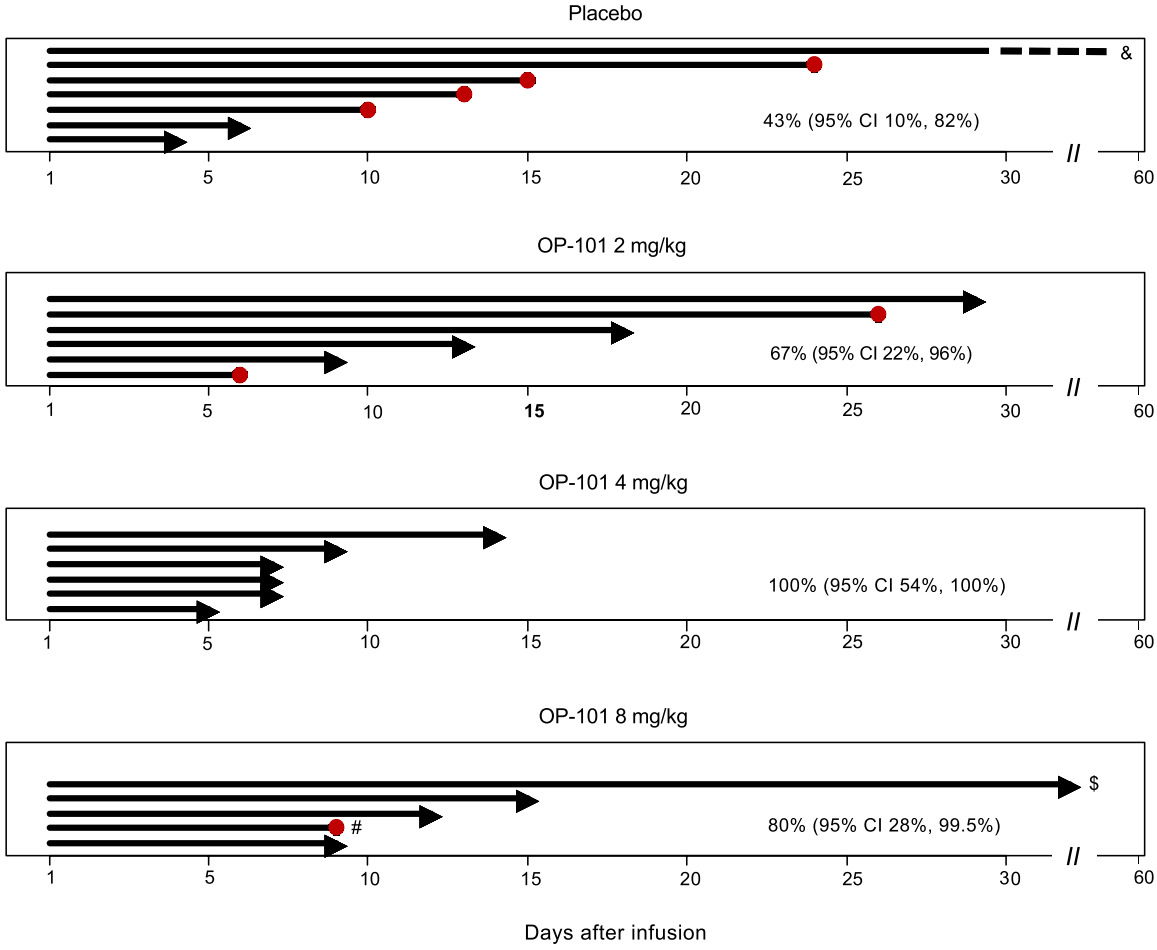


Fig. 6. Time to hospital discharge or death after OP-101 treatment. Shown is the time to discharge from hospital (indicated by an arrowhead) or death (indicated by a red circle) after infusion of placebo or OP-101 at 2, 4, or 8 mg/kg. Percentages and 95% CI indicate fraction of those who survived in each group. &This patient remained in the ICU on mechanical ventilation through tracheostomy until the end of the study at 60 days after infusion. $Patient with chronic liver failure from hepatitis C virus infection. #Patient transitioned to comfort care, and support was withdrawn as per family request. All patients alive at discharge were alive at the last follow-up on day 60 after infusion.

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Table 3. Summary of OP-101 treatment-emergent adverse events. No drug-related adverse events were reported for all doses of OP-101 tested.

Group	OP-101 (2 mg/kg) (*n* = 6) *n* (%)	OP-101 (4 mg/kg) (*n* = 6) *n* (%)	OP-101 (8 mg/kg) (*n* = 5) *n* (%)	Placebo (*n* = 7) *n* (%)	Overall (*n* = 24) *n* (%)
Pharmacokinetic measurements	6 (100%)	6 (100%)	5 (100%)	0	17 (70.8%)
Subjects discontinued from study	2 (33.3%)	0	1 (20.0%)	4 (57.1%)	7 (29.2%)
Adverse event	0	0	0	0	0
Deaths	2 (33.3%)	0	1 (20.0%)	4 (57.1%)	7 (29.2%)
Subjects with at least one of the following:					
TEAE	5 (83.3%)	4 (66.7%)	3 (60.0%)	5 (71.4%)	17 (70.8%)
Drug-related TEAE	0	0	0	0	0
Serious TEAE	4 (66.7%)	2 (33.3%)	2 (40.0%)	4 (57.1%)	12 (50.0%)
Drug-related serious TEAE	0	0	0	0	0

immunosuppression suggests that the bulk of cytokine production may derive from tissue-resident cells, especially macrophages (*54*). The selective localization of OP-101 to activated brain microglia and macrophages in animal models of neuroinflammation (*12*, *13*, *18*, *19*) may enable OP-101 to attenuate tissue-derived inflammation. Uptake of the dendrimer nanoparticles has been shown to be directly proportional to the severity of injury and to the extent of microglial/macropahge activation (*16*). OP-101 may have the potential to improve neurological symptoms by attenuating both systemic and CNS-specific inflammation.

Potent immunomodulators such as tocilizumab (anti–IL-6 receptor monoclonal antibody) and baricitinib (Janus kinase 1 and 2 inhibitor) are currently recommended for use in addition to steroids in critically ill patients with severe COVID-19 (*55*–*57*). At the time that this trial was conducted, these immunomodulators were not routinely recommended; however, future larger trials will be crucial for providing further insights into added benefits of OP-101. The specific cellular targeting seen with OP-101 would enable more precise targeting with minimal off target effects. There is also greater recognition now of the long-term neurological consequences and sequelae of COVID-19 including the cognitive and psychiatric effects (*7*). The effect of OP-101 in attenuating increases in neurological injury markers and potentially neuroinflammation has to be investigated further in larger studies to determine whether this translates into improvements in long-term sequelae.

There are several limitations to our study. The main limitation is the small sample size. This study was designed as a sequential, ascending dose study for preliminary safety evaluation of this nanotherapy. Patients were randomized to placebo or OP-101 with dose escalation to normalize for differences in the disease and disease management over time and to account for variations due to different SARS-CoV-2 strains and patient populations. However, because of the small numbers, it is possible that there was variation/imbalance between the groups. Predose characteristics were similar between the groups except that serum IL-6 concentrations were higher in the patients given placebo. Whereas this may indicate that the placebo group happened to have higher inflammation, we note that this was not the case for other inflammatory markers. The small sample size also precluded correction for potential confounders. Longitudinal changes expressed as rate of change per day assumed linear slopes, and it is likely that changes were nonlinear, but, with a small sample size, we were not powered to explore this. Low increases in analytes are also a limitation in differentiating dose responses. Nonetheless, summarizing changes after treatment with a simple assumption of linear decline is a valid first glimpse of trajectories related to therapy, and future work will need to explore acute and long-term changes in these markers.

This study was not powered for efficacy, and a larger study will be needed to further evaluate safety and efficacy of OP-101 in severe COVID-19. However, this descriptive analysis indicates that the trends seen with inflammatory and neurological injury markers appear to correlate with the trends in improvements in clinical outcomes. We did not have the information from onset of symptoms or diagnosis (polymerase chain reaction positivity) for these patients especially because the recruiting centers were referral centers. Given that there would be variations between time to diagnosis and time to onset of symptoms, we went with a clinically meaningful time point of WHO 7OS of 5 and above to only target patients with severe COVID-19. We believe that these are the patients who had severe inflammation and would benefit most from a targeted therapy.

In conclusion, this study demonstrates that OP-101, a hydroxyl dendrimer conjugated to NAC, shows promising trends in efficacy and safety after a single intravenous dose with improvement in survival and attenuation of inflammatory and neurological injury markers in critically ill patients with COVID-19. As a precision nanomedicine targeting activated macrophages throughout the body, OP-101 has the potential not only to treat the acute inflammatory response in SARS-CoV-2 infection irrespective of the viral variant or viral spike protein mutation but also to treat multiorgan inflammation associated with other disease processes.

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MATERIALS AND METHODS

Study design and participants

This phase 2a clinical study (PRANA) was designed as a multicenter, randomized, double-blind, placebo-controlled, single sequential, ascending-dose study for evaluating safety and tolerability of OP-101 and to obtain preliminary insights on efficacy in patients with severe COVID-19 (ClinicalTrials.gov, NCT04458298). Patients were enrolled at five clinical institutions across the United States: Johns Hopkins School of Medicine, Maryland; Emory University School of Medicine, Georgia; Memorial Hermann Hospital System, University of Texas Health Science Center at Houston, Texas; Broward Health Medical Center, Florida; and Avera McKennan Hospital and University Health Center, South Dakota. Patients with COVID-19 were screened and enrolled within 72 hours because of rapid progression of their disease. Informed consent (English or Spanish) was obtained from the patient or authorized representative. All procedures were approved by a central IRB [Western IRB (WIRB) Copernicus Group (WCG)] and local institutional IRBs and carried out in accordance with the Declaration of Helsinki and principles of Good Clinical Practices. When required, approval was also obtained from the institutional COVID-19 committee.

Inclusion criteria were as follows: male and nonpregnant females aged ≥18 years admitted with a diagnosis of laboratory-confirmed SARS-CoV-2 infection, WHO 7OS score of 5 to 7 (severe COVID-19), body mass index (BMI) of ≤35 kg/m^2, estimated glomerular filtration rate of ≥45 ml/min/1.73 m^2 at screening, and occurrence of at least two of the following vital sign criteria: fever > 38.0°C, tachycardia > 90 beats/min, tachypnea > 20 breaths/min, leukocytosis > 12 × 10^9/liter, or leukopenia < 4 × 10^9/liter; exclusion criteria included patients on extracorporeal membrane oxygenation, expected survival less than 24 hours, neuromuscular disorders, chronic lung disease requiring mechanical ventilation at baseline, dialysis for chronic renal failure, severe liver failure (Childs-Pugh score > 12), myocardial infarction, New York Heart Association class IV congestive heart failure, WHO class III or IV pulmonary hypertension, and active malignancy or recent trauma. Patients receiving other trial immunomodulatory agents were excluded.

Changes to inclusion and exclusion criteria were made on two separate occasions. These changes were done to address lower enrollment rates and to provide clarifications due to delays in transportation to the tertiary enrollment centers and variations in oxygen delivery devices/practices between centers. On 18 November 2020, BMI inclusion was changed to ≤42 kg/m^2; the presence of the vital sign criteria was changed to at any time from the date of hospital admission rather than at screening and to include all delivery modes of oxygen as long as flow rate was >6 liters. On 28 January 2021, the BMI criterion was removed; WHO 7OS score inclusion was clarified to include patients with a score of 4 as long as the flow rate was >6 liters; only one of the vital sign criteria was required; patients requiring HFNC for >72 hours were allowed to be included, however, patients still had to be included within 72 hours of requiring mechanical ventilation. None of the protocol changes affected patient safety or trial objectives. The Food and Drug Administration (FDA)–approved drugs or drugs that were being used under Emergency Use Authorization by the FDA during the period of the trial were allowed. Recruited patients were dosed on day 1, and study follow-up was completed on day 60. Patients in all cohorts received standard-of-care treatment by the primary clinical team, including mechanical ventilation as indicated by clinical judgment and patient response.

There were three separate and sequential cohorts for enrollment with six patients in each treatment group (2, 4, 8 mg/kg OP-101, and placebo). Dose escalation from one cohort to the next higher-dose cohort was determined by the IDMC, after review of safety and tolerability through day 3, with six patients randomized to treatment and two patients to placebo in each enrollment cohort. The first day of enrollment and the last date of follow-up (day 60) for each cohort were 11 August 2020 to 16 November 2020 for cohort 1, 15 October 2020 to 10 March 2021 for cohort 2, and 4 February 2021 to 7 May 2021 for cohort 3. Because this was a proof-of-concept study designed to assess the maximum tolerated dose of OP-101, formal power calculations were not conducted. The sample size of six patients in each group was felt to be adequate to answer broad questions about safety, tolerability, and preliminary efficacy for OP-101 in this critically ill patient population.

Prespecified primary outcome of this stage 1 trial was assessment of safety and tolerability of OP-101 based on the number of participants per group with TEAEs graded, assessed by common terminology criteria for AEs (v4.03). Prespecified secondary outcomes for preliminary efficacy included time to discharge, all-cause mortality at day 30, and percent change in biomarkers over time. Study stop rules included the event of one patient within any dose cohort who had a drug-related serious adverse event (SAE) or ≥2 patients within the same drug cohort with a drug-related grade 3 AE or SAE. Because none of these events occurred, enrollment was not stopped prematurely at any point.

Randomization and blinding

Once enrolled, each patient was randomly allocated to receive either a single intravenous infusion of OP-101 at the enrolling dose or placebo (saline) in a ratio of 3:1 per group using an automated interactive voice and web response system (IXRS). Block randomization (block size of 4) generated with Suvoda Interactive Response Technology (Conshohocken, PA) (third party) was used to assign the patients to treatment or placebo, and a computer-generated serial number was provided by the IXRS based on the schedule assigned to the IXRS. Independent pharmacists at each site were responsible for preparing and dispensing the infusion solution according to the treatment allocation based on the computer-generated serial number via IXRS. Only the independent pharmacists responsible for preparing the solution were unblinded.

All patients, care providers, investigators, site staff involved in assessing safety and efficacy, other related study staff (including the sponsor and any designee), and central laboratories remained blinded to the study treatment assignment throughout the study. Unblinded staff (independent pharmacists and central laboratory staff performing pharmacokinetics) were not involved in any study treatment or assessment. Because OP-101 easily dissolves in saline and is colorless, it is not possible to distinguish between the active and placebo infusion enabling easy maintenance of blinded status for patients and site staff. Data analyses were done after all blinded assessments were completed and once data were locked. The final unblinding of treatments was done after database lock, and data were analyzed by an independent biostatistician not involved with the study.

Study procedures

OP-101 supplied as a lyophilized powder (500 mg per vial) was reconstituted with sterile water and added to a 100-ml intravenous saline bag at the appropriate dose. The 100-ml saline bag (with active

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drug or saline placebo) was administered as a single intravenous infusion over 60 min. After infusion, plasma samples were collected for pharmacokinetic (PK) and pharmacodynamic (PD) analysis. PK samples were obtained with sparse sampling at predose and 2, 12, and 24 hours after infusion. Additional blood samples for PD analysis were taken on days 3, 5, and 8, with safety laboratory tests at screening and on days 1, 2, and 3 and then weekly on days 8, 15, 22, and 30. Urine obtained at these time points was shipped to Nexelis Inc. for KIM-1 and NGAL measurements using standard assay kits. Surviving patients were followed up for 30 days, with a phone follow-up at 60 days, including remote follow-up after discharge.

Safety assessments included AE monitoring, clinical laboratory tests (hematology, chemistry, and urinalysis), physical examinations, and vital signs (heart rate, respiratory rate, body temperature, and blood pressure). Days to discharge or death and days on mechanical ventilation were noted. CRP was measured by the local laboratories. Frozen serum samples collected for PD analysis were analyzed for concentrations of NfL, GFAP, Tau, ubiquitin C-terminal hydrolase L1 (UCH-L1), and various cytokines [interferon-γ (IFN-γ), IL-10, IL-12p70, IL-1β, IL-22, IL-4, IL-5, IL-6, IL-8, and TNF-α] using multiplex immunoassays by central laboratories (Siemens Healthineers and Quanterix). All assays were done blinded and run in duplicate. Values for IFN-γ, IL-12p70, IL-1β, IL-4, IL-5, and UCH-L1 were mostly below the quantification limit; hence, only IL-6, IL-8, IL-10, IL-22, and TNF-α serum concentrations were reported. NfL measured by a clinical laboratory improvement amendments (CLIA)-certified assay and GFAP and Tau protein measured with the 4-plexNeuroA assay were also reported.

Statistical analysis

Medians and proportions summarized continuous and categorical variables and analytic approaches appropriate for small sample sizes were used. To estimate variability around the risk of death by treatment arm, we calculated Clopper-Pearson fiducial 95% CIs (*58*). Fisher's exact test was used to compare differences in proportions by treatment arm. Given that this was a descriptive and preliminary analysis with the goal to identify potential relationships for further investigation, a correction for multiple comparisons to avoid type I error was not necessary, because it would increase the potential for type II error (i.e., incorrectly classifying associations that are not null) (*59*).

To quantify longitudinal changes in markers, we plotted individual data and presented day-specific medians. Predose concentrations between groups were compared using Wilcoxon rank sum test. The analytes were measured longitudinally at standardized times—2, 12, 24 hours, 2, 4, and 7 days after infusion. For each individual, we summarized data by overall trajectories as a slope (i.e., assumption of linear change) by regressing the analyte on time as days. This data reduction technique is useful to summarize individual changes after treatment in this longitudinal setting (*60*), is appropriate for small sample sizes, and allows for analysis with distribution-free nonparametric statistics without violating assumptions of independence. The slope in this simple slope-intercept model for each participant represents the average rate of change in analyte per day. To empirically estimate individual-specific changes, we fit least squares slope-intercept regression models to each patient with the marker as the dependent variable (in the log scale) and time in days from infusion (day 1 or day of infusion) to day 8 (7 days after infusion). When the marker was measured 2 and 12 hours after infusion, this corresponded to 0.083 (= 2 of 24) and 0.5 (= 12 of 24) days in the model. This model was of the form

$$\log(\text{biomarker}_{ij}) = a_i + b_i t_{ij}$$

with i indexing the individual and j indexing the time. Here, $\exp(a_i)$ is the estimated marker for individual i at $t_{i1} = 0$ (or time of infusion), and b_i is the change per day in log-transformed marker for the ith individual. This coefficient can be expressed as percent change per day by calculating $(\exp(b_i) - 1) \times 100$. To compare empirical estimates of longitudinal changes by treatment arm, pair-wise Wilcoxon rank sum test was used with placebo as the reference to test for differences across treatment arms. Given the small sample size and exploratory nature of this analysis and to limit type II error, statistical significance was evaluated at $\alpha = 0.05$ (*59*). Statistical analyses were conducted in SAS 9.4 (SAS Institute Inc., Cary, North Carolina), and R version 4.0.0. GraphPad Prism 9.0 was used to plot the graphs.

SUPPLEMENTARY MATERIALS

www.science.org/doi/10.1126/scitranslmed.abo2652
Figs. S1 to S7
Tables S1 to S4
Data file S1

View/request a protocol for this paper from *Bio-protocol*.

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Dendrimer nanotherapy for severe COVID-19 attenuates inflammation and neurological injury markers and improves outcomes in a phase2a clinical trial

Aaron M. Gusdon, Nauder Faraday, John S. Aita, Sunil Kumar, Ishan Mehta, HuiMahn A. Choi, Jeffery L. Cleland, Keith Robinson, Louise D. McCullough, Derek K. Ng, Rangaramanujam M. Kannan, and Sujatha Kannan

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A dendrimer nanotherapy for COVID-19

OP-101, a hydroxyl-polyamidoamine dendrimer–*N*-acetyl cysteine conjugate, that specifically targets activated macrophages was evaluated in patients with severe COVID-19 in a phase 2a clinical trial. A single intravenous infusion of OP-101 was well tolerated and reduced the risk of death or mechanical ventilation at 30 and 60 days after treatment when compared to placebo. OP-101 treatment was associated with decreased serum concentrations of proinflammatory markers as well as markers of neurological injury (neurofilament light chain and glial fibrillary acidic protein). OP-101 may have potential for treating systemic hyperinflammation in severe COVID-19.

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